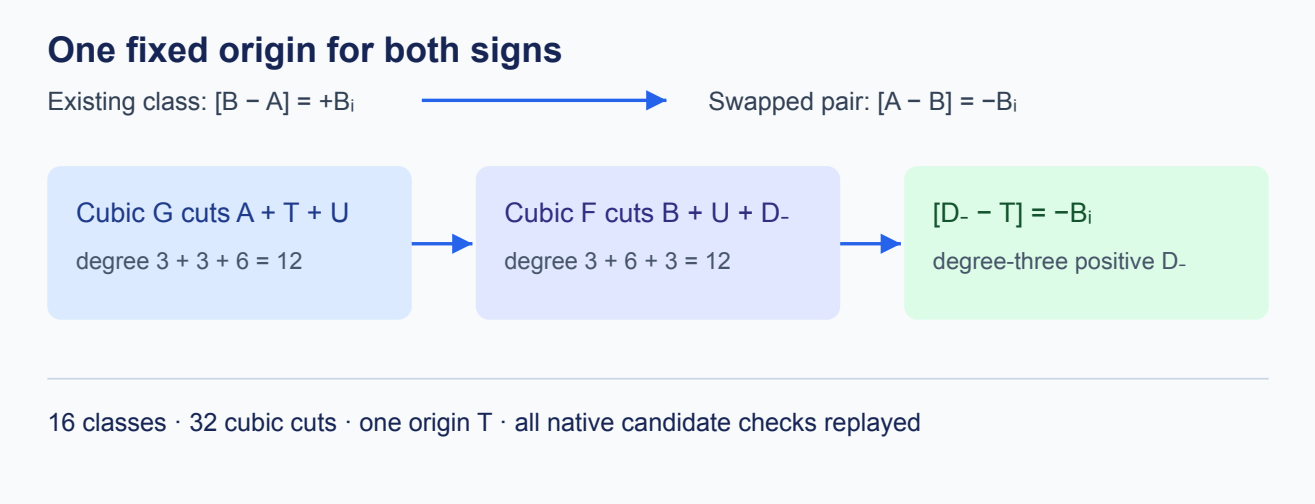
P-192 Type I negative classes at one fixed origin

All 16 negatives of the previously reduced Type I base classes now have positive degree-three divisor representatives relative to the same origin T. Swapping the old input pair changes its class from $[B - A]$ to $[A - B]$. Native PARI constructs two cubic sections per class, and a separate checker replays all 32 cuts, residual coordinates, squarefreeness, disjointness, and rank-nine interpolation.



Class identity

For the swapped pair, cubic G cuts $A + T + U$ with $\deg U = 6$. Cubic F cuts $B + U + D_-$ with $\deg D_- = 3$. Their common line-bundle class implies $[D_- - T] = [A - B] = -[B - A]$. Lean checks this implication in `negative_fixed_origin_cubic_class_identity` under the two section-divisor equalities. The concrete P-192 quartic and sections are verified by native finite-field arithmetic.

Exact replay

The frozen input contains the quartic, three anchor points, and 16 original degree-three pairs. Each negative reduction succeeds with the first denominator trial $[1, 1, 1, 1]$. The producer writes all cubic coefficients and triangular $\frac{x}{y}$ residual polynomials. The separate verifier recomputes both resultants and the unique numerator cubic, ending with `NEGATIVE_FIXED_ORIGIN_VERIFY_PASS classes=16 cuts=32 prime_bits=192`.

This construction makes the 32 signed classes available at one fixed origin. It does not change the earlier fixed-base three-sum support or the 64-query zero-rank result. The next collector must measure ordinary-query yield, useful rank, and signed-index cost on a larger frozen workload.

Exact input and source hashes, commands, proof boundary, and links: `research/p192_extension_cover_20261012/reports/run-20261015T1724Z/REPORT.md`.